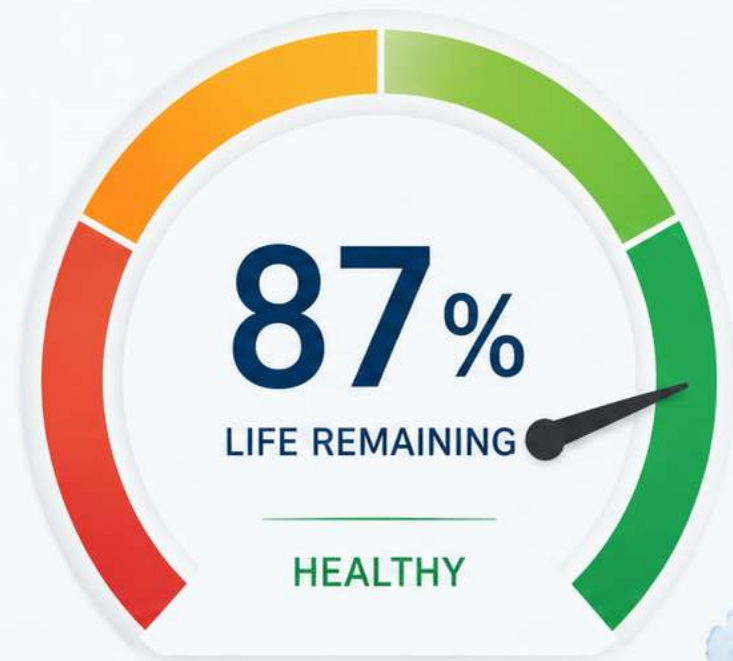


トランスの じゅみょう メーター Transformer Life Gauge

Real-Time Estimation of Transformer Insulation Life



Turning transformer thermal behavior into an easy-to-understand health indicator.

A presentation by
Rahul Sahu
Aditya University

トランスの じょうたい が わかる

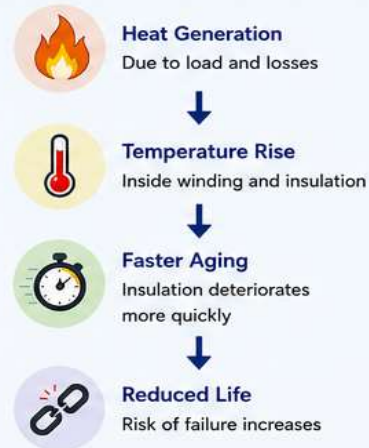
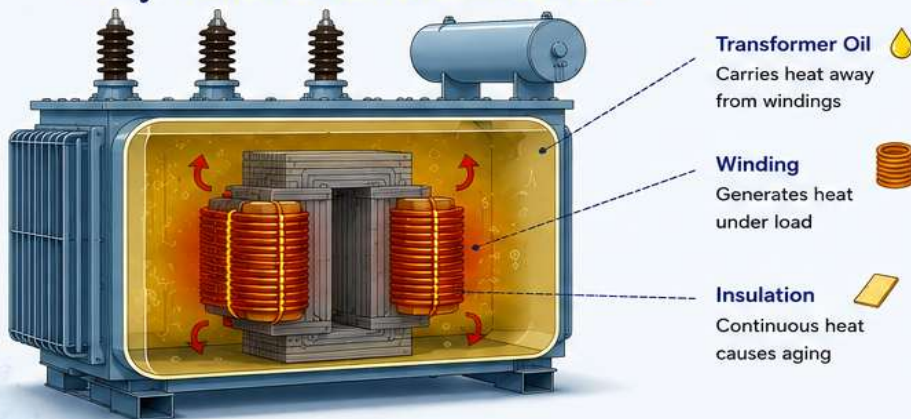
いまの ほしゅ と じゅみょうの わけ Why Current Maintenance Leads to Life Decrease

Maintenance is time-based, not condition-based. Heat is the main reason for insulation aging.



Maintenance happens at fixed intervals, not based on the real condition of the transformer.

Why Transformer Life Decreases



Industry Failure Data

Source: CIGRE Transformer Reliability Survey



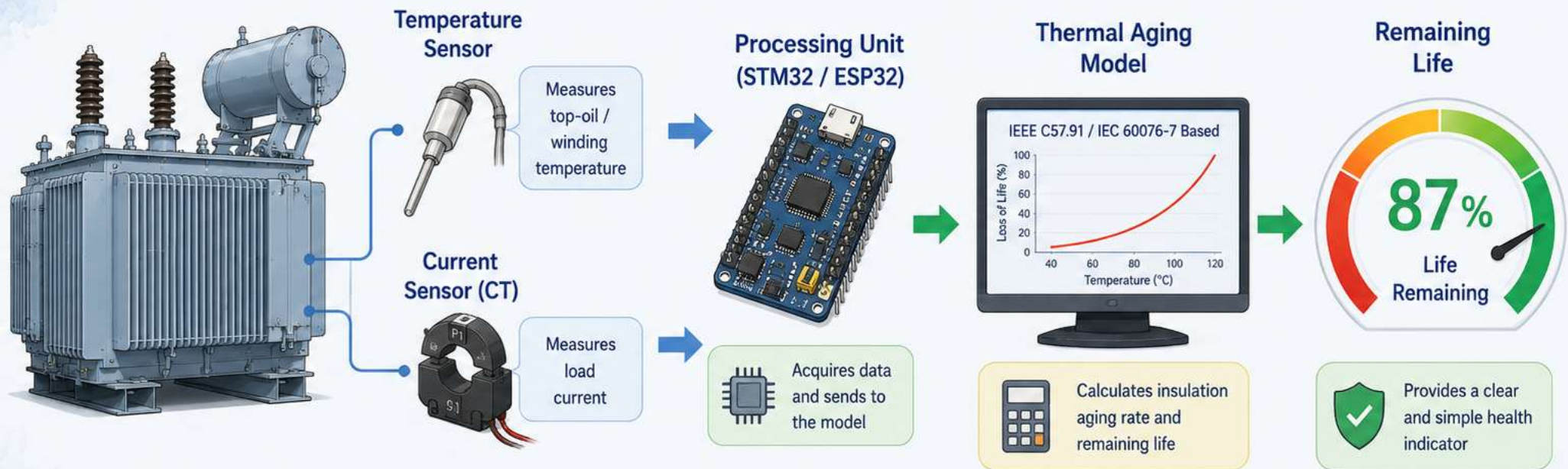
25% of known failure causes come from insulation ageing + improper maintenance.

~0.3–0.4% / year
Overall large power transformer failure rate (CIGRE, modern units).

あたらしい アイデア

Transformer Life Gauge Concept

Use key operating parameters and a thermal aging model to estimate remaining insulation life.



- ✓ Combines temperature and load current for accurate life estimation.
- ✓ Based on industry-standard thermal aging principles.
- ✓ Enables real-time assessment of transformer insulation health.





システムのこうせい Life Calculation Model



Three-step mathematical model to estimate transformer life in real-time.



MATLAB Offline Simulation (24-Hour Example)



Physics-Based Model
(IEEE C57.91)



Accurate Real-Time
Life Estimation



Smarter Loading
Decisions



Extends Transformer
Life & Reliability



Offline MATLAB Validation

The mathematical model is validated with a simulated daily load profile before deployment on the IoT hardware.

ライブ モニター Live Monitoring Dashboard

Real-time data visualization and instant alerts for better decision making.

モニター できるもの What We Monitor



おんど
Oil / Winding
Temperature



でんりゅう
Load Current



かんきょう
おんど
Ambient
Temperature



うんてん じかん
Operating Hours



どこからでも
かんたんに チェック!
Easy to check
from anywhere!



ひとめで わかる
Easy to Understand
at a Glance



リアルタイム データ
Real-Time Data
Updates



アラートで まえに きづく
Early Alerts,
Prevent Failures



データで よりよい
かんりを サポート
Data-Driven
Better Management

まとめとこれから

Conclusion & Future Scope

Smarter monitoring today, stronger power tomorrow.

WHAT WE ACHIEVED



FUTURE SCOPE



あんぜんな データつうしん
Secure Data Transmission



リアルタイム モニター
Real-Time Monitoring



アラートで じこを よぼう
Early Fault Detection



データで じゅみようを のぼす
Data-Driven Life Extension



ありがとう
ございます

Thank You

Your time and attention are
highly appreciated.



Let's build a smarter
and safer power future,
together.



Reliable
Monitoring



Instant
Alerts



Smart
Insights



Better
Power Future